



PRE-DSA FOUNDATIONS (Must Do Before LeetCode)

0 Fundamentals of Programming

Programming Basics

- What is Programming?
 - Flowcharts
 - Logical Thinking & Problem Solving
-

1 Language Basics (Choose ONE main language)

C++

- C++ Setup
- C++ Basics
- STL Basics

Java

- Java Setup
- Java Basics
- Java OOPs Basics
- Java Collections Framework

Python

- Python Setup
- Python Basics
- Python Libraries – Part 1
- Python Libraries – Part 2

 *Pick ONE language for DSA (C++ / Java recommended). Python is okay but slower in contests.*

2 Input / Output & Control Flow

Input / Output

- Input Output Basics
- Print X N times
- Print last character of string

Conditional Statements

- If–Else (Adult / Teen problem)
 - If–Else If ladder
 - Switch Case
-

3 Basic Problem Solving (Warm-up)

- Sum of first and last element in array
 - Print X N numbers of times
 - Print last character of string
-

4 Logic Building – Patterns

Easy → Medium → Hard (in order)

- Logic Building (Patterns – Easy)
- Logic Building (Patterns – Medium)
- Logic Building (Patterns – Hard)

Pattern Problems

- Pattern 1
- Pattern 2
- Pattern 3
- Pattern 4
- Pattern 5
- Pattern 6
- Pattern 7
- Pattern 8
- Pattern 9
- Pattern 10
- Pattern 11
- Pattern 12
- Pattern 13
- Pattern 14
- Pattern 15
- Pattern 16
- Pattern 17

- Pattern 18
- Pattern 19
- Pattern 20
- Pattern 21
- Pattern 22

 *Patterns build loops, conditions, and confidence. Do NOT skip.*

5 Time & Space Complexity

Theory

- What is Time Complexity?
 - Big-O, Big-Omega, Big-Theta
 - Best / Average / Worst Case
 - Time Complexity with Examples
 - Space Complexity Basics
-

6 Concept Basics (Core Foundations)

Overview

- Basic Maths
 - Basic Arrays
 - Basic Hashing
 - Basic Strings
-

7 Basic Maths (Very Important)

- Count all digits of a number
 - Count odd digits in a number
 - Reverse a number
 - Palindrome number
 - Largest digit in a number
 - Factorial of a number
 - Armstrong number
 - Perfect number
 - Check prime number
 - Count primes till N
 - GCD of two numbers
 - LCM of two numbers
 - Print divisors of a number
-

8 Basic Arrays

- Sum of array elements
 - Count odd numbers in array
 - Check if array is sorted I
 - Reverse an array
-

9 Basic Hashing

- Highest occurring element in array
 - Second highest occurring element
 - Sum of highest and lowest frequency
-

10 Basic Strings

- Reverse a string II
 - Palindrome check
 - Largest odd number in a string
 - Longest common prefix
 - Isomorphic string
 - Rotate string
 - Valid anagram
 - Sort characters by frequency
-

11 Basic Recursion

Theory

- What is Recursion?
- Recursion Tree
- Parameters & Base Case

Problems

- Sum of first N numbers
- Factorial of a number

- Sum of array elements
 - Reverse a string I
 - Check palindrome string
 - Check prime using recursion
 - Reverse an array
 - Check if array is sorted II
 - Sum of digits of a number
 - Fibonacci number
-

Transition Point (VERY IMPORTANT)

Once you finish **Basic Recursion**, you are officially **DSA-ready** 🚀
From here you jump directly into:

👉 **Sorting** → **Arrays** → **Hashing** → **Binary Search** → ... (the roadmap I gave earlier)

Suggested Order (One-Line Cheat Sheet)

Programming → Language → I/O → If-Else → Loops → Patterns
→ Time Complexity → Maths → Arrays → Hashing → Strings → Recursion →
Sorting → DSA Proper → LeetCode

LeetCode DSA Learning Roadmap (Progressive Order)

Sorting Algorithms

Basics

- Selection Sort
- Bubble Sort
- Insertion Sort

Efficient Sorting

- Merge Sort
 - Quick Sort
-

2 Arrays

Fundamentals

- Linear Search
- Largest Element
- Second Largest Element
- Maximum Consecutive Ones
- Left Rotate Array by One
- Left Rotate Array by K Places

Logic Building

- Move Zeros to End
- Remove duplicates from sorted array
- Find missing number
- Union of two sorted arrays
- Intersection of two sorted arrays

FAQs (Medium)

- Majority Element I
- Leaders in an Array
- Rearrange array elements by sign
- Print the matrix in spiral manner
- Pascal's Triangle I
- Pascal's Triangle II
- Pascal's Triangle III
- Rotate matrix by 90 degrees
- Two Sum
- Three Sum
- Four Sum
- Sort an array of 0s, 1s and 2s
- Kadane's Algorithm
- Next Permutation

FAQs (Hard)

- Majority Element II
- Find the repeating and missing number
- Count Inversions
- Reverse Pairs
- Maximum Product Subarray
- Merge two sorted arrays without extra space

3 Hashing

Theory

- Basic Hashing Concepts

Problems

- Longest Consecutive Sequence
 - Longest Subarray with Sum K
 - Count subarrays with given sum
 - Count subarrays with given XOR K
-

4 Binary Search

Fundamentals

- Search X in sorted array
- Lower Bound
- Upper Bound

Logic Building

- Search Insert Position
- Floor and Ceil in sorted array
- First and last occurrence
- Search in rotated sorted array I
- Search in rotated sorted array II
- Find minimum in rotated sorted array
- Number of rotations in sorted array
- Single element in sorted array

Binary Search on Answers

- Square root of a number
- Nth root of a number
- Find smallest divisor
- Koko Eating Bananas
- Minimum days to make M bouquets

FAQs

- Aggressive Cows
 - Book Allocation Problem
 - Find Peak Element
 - Median of two sorted arrays
 - Kth element of two sorted arrays
 - Minimize maximum distance to gas station
 - Split Array – Largest Sum
-

5 2D Arrays / Matrices

- Find row with maximum 1s
 - Search in a 2D matrix
 - Search in 2D matrix II
 - Find Peak Element II
 - Matrix Median
-

6 Recursion & Backtracking

Basics

- Pow(x, n)
- Generate Parentheses
- Power Set

Subsequence Pattern

- Check if subsequence with sum K exists
- Count all subsequences with sum K

FAQs (Medium)

- Combination Sum
- Combination Sum II
- Subsets I
- Subsets II
- Combination Sum III

Hard

- Letter Combinations of Phone Number

FAQs (Hard)

- Palindrome Partitioning
 - Word Search
 - N Queen
 - Rat in a Maze
 - M Coloring Problem
 - Sudoku Solver
-

7 Linked List

Singly Linked List (Fundamentals)

- Introduction
- Traversal
- Insertion
- Deletion
- Delete head
- Delete tail
- Delete Kth node
- Delete node with value X
- Insert at head
- Insert at tail
- Insert at Kth position
- Insert before value X

Doubly Linked List

- Introduction
- Insertion
- Deletion
- Convert array to DLL
- Delete head
- Delete tail
- Delete Kth node
- Remove given node
- Insert before head

- Insert before tail
- Insert before Kth node
- Insert before given node

Logic Building

- Add two numbers
- Segregate odd and even nodes
- Sort LL of 0s, 1s and 2s
- Remove Nth node from end
- Reverse Linked List

FAQs (Medium)

- Add one to number represented by LL
- Find middle
- Delete middle
- Check palindrome
- Intersection point of Y LL
- Detect loop
- Starting point of loop
- Length of loop

FAQs (Hard)

- Reverse LL in groups of K
- Rotate Linked List
- Merge two sorted lists
- Flatten Linked List

- Sort Linked List
 - Clone LL with random pointer
-

8 Bit Manipulation

Theory

- Bits & Tricks

Problems

- Minimum bit flips
 - Single Number I
 - Single Number II
 - Single Number III
 - Divide two numbers without *, /
 - Power set using bits
 - XOR of numbers in range
-

9 Greedy Algorithms

Easy

- Assign Cookies
- Lemonade Change
- Jump Game I

Scheduling / Intervals

- Shortest Job First

- Job Sequencing
- N meetings in one room
- Non-overlapping intervals
- Insert Interval
- Minimum platforms

Hard

- Valid Parenthesis Checker
 - Candy
-

10 Sliding Window & Two Pointers

Theory

- Sliding Window Patterns

Constant Window

- Maximum points from cards

Variable Window

- Longest substring without repeating characters
- Max consecutive ones III
- Fruit into baskets
- Longest substring with at most K distinct characters
- Longest repeating character replacement
- Minimum window substring

Counting Subarrays

- Substrings containing all three characters
 - Binary subarrays with sum
 - Count nice subarrays
-

Stack & Queue

Implementation

- Stack using array
- Queue using array
- Stack using queue
- Queue using stack
- Stack using linked list
- Queue using linked list
- Balanced parentheses

Monotonic Stack

- Next greater element
- Next greater element II
- Asteroid collision
- Sum of subarray minimums
- Sum of subarray ranges
- Remove K digits

FAQs

- Min Stack
- Sliding Window Maximum

- Trapping Rainwater
 - Largest Rectangle in Histogram
 - Maximum Rectangle
 - Stock Span
 - Celebrity Problem
 - LRU Cache
 - LFU Cache
-

Binary Trees

Traversals

- Inorder
- Preorder
- Postorder
- Level Order
- All traversals in one

Medium

- Max depth
- Identical trees
- Balanced tree
- Diameter
- Maximum path sum
- Symmetric tree

FAQs

- Zigzag traversal
- Boundary traversal
- Vertical traversal
- Top / Bottom view
- Left / Right view
- Root to node path
- LCA
- Maximum width
- Nodes at distance K
- Burning tree
- Count nodes in complete tree

Construction

- Build from preorder + inorder
- Build from postorder + inorder
- Serialize & deserialize

Constant Space

- Morris inorder
- Morris preorder

Binary Search Trees

- Search
- Insert
- Delete

- Kth smallest / largest
- Validate BST
- LCA
- Construct from preorder
- Inorder successor & predecessor

FAQs:

- BST Iterator
- Two Sum in BST
- Recover BST
- Largest BST in BT

14 Heaps

Theory

- Heap concepts
- Heapify
- Build heap
- Min heap
- Max heap
- Heap sort
- Kth largest element

FAQs

- Kth largest in stream
-

15 Graphs

Basics

- Graph intro
- BFS & DFS
- Connected components

Traversal Problems

- Provinces
- Islands
- Flood fill
- Enclaves
- Rotten oranges
- Nearest 1
- Surrounded regions
- Distinct islands

Cycles & Topo

- Cycle detection (undirected)
- Bipartite graph
- Topological sort
- Cycle in directed graph

Hard

- Eventual safe states
- Course schedule I & II
- Alien dictionary

- Shortest path DAG
- Shortest path unit weight
- Word ladder I & II

Shortest Path

- Dijkstra
- Bellman Ford
- Floyd Warshall
- Binary maze
- Path with minimum effort
- Cheapest flights
- Number of ways to arrive

MST

- MST theory
- Disjoint Set
- MST weight

Advanced

- Kosaraju
- Bridges
- Articulation points

Dynamic Programming

Basics

- DP Introduction

1D DP

- Climbing stairs
- Frog jump
- Frog jump K
- Max sum non-adjacent
- House robber

2D / Grid DP

- Ninja training
- Unique paths
- Unique paths II
- Falling path sum
- Triangle
- Cherry pickup II

DP on Stocks

- Buy & sell I – IV
- With transaction fee

DP on Subsequences

- Subset sum
- Partition equal subset
- Min difference partition
- Count subsets

- Knapsack
- Coin change
- Target sum
- Rod cutting

LIS

- LIS
- Print LIS
- Longest divisible subset
- Longest string chain
- Bitonic subsequence
- Number of LIS

DP on Strings

- LCS
- Longest common substring
- Longest palindromic subsequence
- Edit distance
- Wildcard matching
- Shortest supersequence

MCM DP

- Matrix chain multiplication
 - Burst balloons
 - Palindrome partitioning II
-

17 Tries

- Trie implementation
 - Advanced trie ops
 - Longest word with all prefixes
 - Distinct substrings
 - Maximum XOR
 - Maximum XOR with constraints
-

18 Strings (Advanced Algorithms)

- Reverse words
- Minimum bracket reversals
- Count and say

Advanced

- Rabin Karp
 - Z Algorithm
 - KMP / LPS
 - Shortest palindrome
 - Longest happy prefix
-

19 Maths

- Sieve of Eratosthenes
- Print primes till N

- Prime factorization
 - Count primes in range L to R
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WHEN TO SWITCH: Learning → Solving (RULES)

The Golden Rule

For **each topic**:

- Learn concept + 1–2 examples
- Solve **8–12 problems**
- If you can solve **60–70% without seeing solution** → MOVE ON

Do NOT:

- Watch 10 videos before solving
- Memorize solutions
- Stay stuck on one topic for weeks

Time Rule

- **Easy**: 15–25 min
- **Medium**: 30–45 min
- **Hard**: skip initially, come back later

If stuck:

- Hint → partial idea → full solution (only after 30–45 min)
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PRE-DSA / BASICS (LeetCode)

Basic Maths

Solve after learning loops & conditionals.

- LC 7 – Reverse Integer
- LC 9 – Palindrome Number
- LC 231 – Power of Two
- LC 326 – Power of Three
- LC 342 – Power of Four
- LC 202 – Happy Number
- LC 258 – Add Digits
- LC 628 – Maximum Product of Three Numbers

→ **Switch after:** you stop fearing numbers & modulo logic

Basic Arrays

- LC 1480 – Running Sum of 1D Array
- LC 724 – Find Pivot Index
- LC 977 – Squares of a Sorted Array
- LC 1929 – Concatenation of Array
- LC 217 – Contains Duplicate
- LC 53 – Maximum Subarray (Kadane intro)

→ **Switch after:** loops feel natural

Basic Strings

- LC 344 – Reverse String
- LC 125 – Valid Palindrome
- LC 242 – Valid Anagram
- LC 58 – Length of Last Word
- LC 14 – Longest Common Prefix
- LC 205 – Isomorphic Strings

→ **Switch after:** comfortable with indices & conditions

Basic Recursion

- LC 509 – Fibonacci Number
- LC 50 – Pow(x, n)
- LC 206 – Reverse Linked List (recursive)
- LC 344 – Reverse String (recursive version)

→ **Switch after:** base case instinctively clicks



CORE DSA (VERY IMPORTANT)

Sorting

Learn → immediately solve.

- LC 912 – Sort an Array
- LC 75 – Sort Colors
- LC 169 – Majority Element
- LC 215 – Kth Largest Element

→ **Switch after:** you know when to use merge vs quick vs counting

Arrays (Core)

- LC 1 – Two Sum
- LC 121 – Best Time to Buy and Sell Stock
- LC 122 – Buy & Sell Stock II
- LC 189 – Rotate Array
- LC 283 – Move Zeroes
- LC 268 – Missing Number
- LC 485 – Max Consecutive Ones

→ **Switch after:** you think in terms of indices, not brute force

Hashing

- LC 217 – Contains Duplicate
- LC 242 – Valid Anagram
- LC 560 – Subarray Sum Equals K
- LC 128 – Longest Consecutive Sequence
- LC 15 – 3Sum

→ **Switch after:** map/set feels automatic

Binary Search

- LC 704 – Binary Search
- LC 35 – Search Insert Position

- LC 34 – First & Last Position
- LC 153 – Find Min in Rotated Sorted Array
- LC 33 – Search in Rotated Sorted Array
- LC 69 – Sqrt(x)
- LC 875 – Koko Eating Bananas

→ **Switch after:** you can write binary search without template

Sliding Window / Two Pointers

- LC 3 – Longest Substring Without Repeating Characters
- LC 424 – Longest Repeating Character Replacement
- LC 1004 – Max Consecutive Ones III
- LC 904 – Fruit Into Baskets
- LC 76 – Minimum Window Substring

→ **Switch after:** window expansion/shrinking is intuitive

Stack / Queue

- LC 20 – Valid Parentheses
- LC 155 – Min Stack
- LC 496 – Next Greater Element I
- LC 503 – Next Greater Element II
- LC 84 – Largest Rectangle in Histogram
- LC 42 – Trapping Rain Water

→ **Switch after:** monotonic stack “clicks”

Linked List

- LC 206 – Reverse Linked List
- LC 876 – Middle of Linked List
- LC 141 – Linked List Cycle
- LC 142 – Cycle II
- LC 234 – Palindrome Linked List
- LC 19 – Remove Nth Node
- LC 160 – Intersection of Two LL

➡ **Switch after:** pointer movement feels natural

Recursion / Backtracking

- LC 78 – Subsets
- LC 90 – Subsets II
- LC 39 – Combination Sum
- LC 40 – Combination Sum II
- LC 46 – Permutations
- LC 51 – N Queens

➡ **Switch after:** you can visualize recursion tree

Binary Trees

- LC 94 – Inorder Traversal
- LC 102 – Level Order Traversal

- LC 104 – Max Depth
- LC 110 – Balanced Binary Tree
- LC 543 – Diameter of Binary Tree
- LC 124 – Maximum Path Sum
- LC 236 – LCA

➡ **Switch after:** traversal choice is obvious

BST

- LC 700 – Search in BST
 - LC 701 – Insert into BST
 - LC 98 – Validate BST
 - LC 230 – Kth Smallest
 - LC 653 – Two Sum IV
-

Heaps

- LC 215 – Kth Largest Element
 - LC 347 – Top K Frequent Elements
 - LC 703 – Kth Largest in Stream
 - LC 295 – Find Median from Data Stream
-

Graphs

Start ONLY after trees.

- LC 200 – Number of Islands
- LC 733 – Flood Fill
- LC 994 – Rotten Oranges
- LC 207 – Course Schedule
- LC 210 – Course Schedule II
- LC 127 – Word Ladder
- LC 785 – Bipartite Graph

→ **Switch after:** BFS vs DFS decision is instant

Dynamic Programming

DO NOT rush DP.

Start with:

- LC 70 – Climbing Stairs
- LC 198 – House Robber
- LC 213 – House Robber II
- LC 322 – Coin Change
- LC 416 – Partition Equal Subset Sum
- LC 1143 – LCS

Then:

- LC 300 – LIS
- LC 72 – Edit Distance
- LC 115 – Distinct Subsequences

→ **Switch after:** you can define state + transition yourself

Learning : Solving = 30 : 70

Revisit topics after 2–3 weeks (revision is gold)